Active Brownian Particles in a Magnetic Field

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1 Introduction

The Bohr-van Leeuwen(BvL) theorem[1][2] is a classic result of physics, that proves the absence of diamagnetic moment for a classical system of charged particles in a constant magnetic field, at thermal equilibrium. It basically put an end to any attempt at explaining the microscopic origin of magnetism using just classical and statistical mechanics.

This fact might appear counter-intuitive because in presence of magnetic field, it is known that a charged particle revolves in a circular orbit, with frequency of orbit $= \omega_c$, the cyclotron frequency. One might be led to conclude, then, that at equilibrium, the charged Brownian particles should also behave in a similar way, producing some kind of current loop(s).

However, the catch is that a system without any confining boundary can never have an equilibrium distribution, since the particle's space coordinate is capable of becoming unbounded. Then we are led to consider a finite system,

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e.g. a square region, with rigid walls (which can be modeled with a steeply rising wall potential). The bulk can have current loops as intuitively expected; the boundaries, however, have skipping orbits as shown in Figure 1. The surprising fact is that these skipping orbits carry exactly equal current as the loops in the bulk, though in an opposite sense. These currents cancel, leading to the validity of the aforementioned theorem. This result holds for any confining potential, regardless of its details.

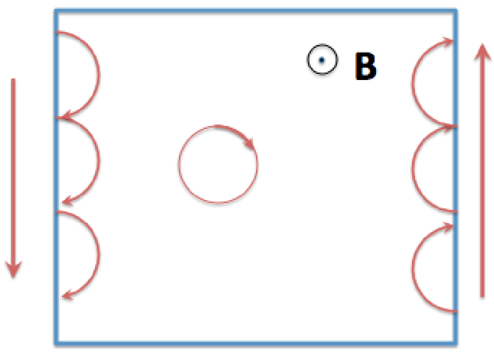

Figure 1: Graphical illustration of BvL theorem [3]

In this report, we study a minimally out-of-equilibrium extension of the system described above. In our system described earlier, we now introduce an exponentially correlated Gaussian noise term (also referred to as coloured noise). We also put in a confining potential, and exactly solve for the probability distribution function of the position of the particle in the case where this potential is simple harmonic. In case of a generalised confining potential, we attempt to quantify the non-equilibrium behaviour in terms of a generalised fluctuation dissipation theorem akin to [4].

2 Equations of motion

Using Newtonian mechanics

$$\dot{\boldsymbol{x}} = \boldsymbol{v} \tag{1}$$

$$m\dot{\boldsymbol{v}} = -\Gamma \boldsymbol{v} + q\boldsymbol{v} \times \boldsymbol{B} + \boldsymbol{F} + \boldsymbol{\eta}(t) \tag{2}$$

Here $\boldsymbol{\eta}(t)$ is a coloured noise term having the following statistical properties

$$\langle \eta_i(t) \rangle = 0$$

$$\langle \eta_i(t)\eta_j(t') \rangle = \frac{A}{\tau}\delta_{ij} \exp \frac{|t-t'|}{\tau}$$

These statistical properties of the coloured noise can be modeled by imposing the following dynamical equation for $\boldsymbol{\eta}(t)$

$$\tau\dot{\boldsymbol{\eta}} = -\boldsymbol{\eta} + \sqrt{2A}\boldsymbol{\zeta}(t) \tag{3}$$

where $\boldsymbol{\zeta}$ is unit strength Gaussian white niose. It should be noted that, in the limit of $\tau \to 0$, $\boldsymbol{\eta}(t)$ becomes a white noise term.

In colloid science and cell biology, inertia plays a negligible role, so that it is appropriate to ignore inertia. Since we are going to look at our problem in this over-damped limit, we can drop the term on the LHS of (2) in order to get a dynamical equation in $\boldsymbol{x}$ only:

$$\Gamma\dot{\boldsymbol{x}} - q\dot{\boldsymbol{x}} \times \boldsymbol{B} = \boldsymbol{F} + \boldsymbol{\eta}(t) \tag{4}$$

By defining $\mathbb{M} = \begin{pmatrix} \Gamma & -qB \\ qB & \Gamma \end{pmatrix}^{-1}$, we can re-write (4) as follows:

$$\mathbb{M}^{-1}\dot{\boldsymbol{x}} = \boldsymbol{F} + \boldsymbol{\eta}(t) \tag{5}$$

Using (3), we can reduce (5) to the following pair of equations:

$$\dot{\boldsymbol{x}} = \mathbb{M}\boldsymbol{u} \tag{6}$$

$$\tau\dot{\boldsymbol{u}} + \boldsymbol{u} = \boldsymbol{F} + \tau(\mathbb{M}\boldsymbol{u} \cdot \nabla)\boldsymbol{F} + \sqrt{2A}\boldsymbol{\zeta}(t) \tag{7}$$

where we have introduced a new velocity-like variable $\boldsymbol{u} = \mathbb{M}^{-1}\dot{\boldsymbol{x}}$.

It is interesting to note in (7) that τ appears where a mass would normally appear. Note that the process of expressing the coloured noise $\boldsymbol{\eta}$ in terms of the white noise $\boldsymbol{\zeta}$ via (3) has resulted in the appearance of an effective inertia in the form of τ.

Another point that needs to be stressed here is the fact that equations (5)-(7) are not only applicable to charged particles in a magnetic field, but also to spinning (uncharged) classical particles[5]. The non-zero off-diagonal elements of the mobility matrix $\mathbb{M}$ are the signature of swerving motion, an example of which is the Magnus force on spinning particles. So the results derived in this report could as well be applied to such systems of spinning particles, albeit with relevant mapping of the parameters.

3 Solution for harmonic potential

Equations (6) and (7) can be re-written as follows for $\boldsymbol{F} = -k\boldsymbol{x}$:

$$\dot{\boldsymbol{x}} = \mathbb{M}\boldsymbol{u} \tag{8}$$

$$\dot{\boldsymbol{u}} = -\frac{1}{\tau}\boldsymbol{u} - \frac{k}{\tau}\boldsymbol{x} - k\mathbb{M}\boldsymbol{u} + \frac{\sqrt{2A}}{\tau}\boldsymbol{\zeta}(t) \tag{9}$$

Let $\boldsymbol{\lambda} = \{\boldsymbol{x}, \boldsymbol{u}\}^T$, we can write the system of Langevin equations as:

$$\dot{\boldsymbol{\lambda}} = \mathbb{A}\boldsymbol{\lambda} + \frac{\sqrt{2A}}{\tau}\begin{pmatrix} \mathbf{0} \\ \boldsymbol{\zeta}(t) \end{pmatrix}$$

where $\mathbb{A} = \begin{pmatrix} \mathbb{O} & \mathbb{M} \\ -\frac{k}{\tau}\mathbb{I} & -\left(\frac{1}{\tau}\mathbb{I} + k\mathbb{M}\right) \end{pmatrix}$ with $\mathbb{O}$ representing the null 2x2 matrix.

Solving the above Langevin equations by standard techniques [6], we arrive at the following probability distribution function in configuration space:

$$\frac{\Gamma k\left(B^2q^2 + (\Gamma + k\tau)^2\right)\exp\left(-\frac{\Gamma k\left(x^2+y^2\right)\left(B^2q^2+(\Gamma+k\tau)^2\right)}{2A\left(B^2q^2+\Gamma(\Gamma+k\tau)\right)}\right)}{2\pi A\left(B^2q^2+\Gamma(\Gamma+k\tau)\right)}$$

where $A = k_BT\Gamma$.

We notice that taking the limit of $B = 0$, we get back the same probability distribution as derived by Joanny et. al.[7]

Diamagnetic moment, defined as $\frac{q}{2}\langle\boldsymbol{x}\times\dot{\boldsymbol{x}}\rangle$, is found to be $\frac{ABq^2}{\Gamma\left(B^2q^2+(\Gamma+k\tau)^2\right)}$ which, for the limit $\tau \approx 0$ reduces to $\frac{ABq^2}{\Gamma\left(B^2q^2+\Gamma^2\right)}$, exactly what is expected in the inertia-less equilibrium limit.

It is also interesting to note that the effect of the magnetic field is only seen in the above-mentioned probability distribution function when τ is non-zero. In other words, the effect of the magnetic field vanishes for the equilibrium case of $\tau = 0$, reducing the probability distribution function to the ordinary canonical distribution.

4 The general problem

For a potential Φ, the equations of motion are written as:

$$\dot{\boldsymbol{x}} = \boldsymbol{v} \tag{10}$$

$$\tau\dot{\boldsymbol{v}} = -\tau(v_k\partial_k)\mathbb{M}\nabla\Phi - \boldsymbol{v} - \mathbb{M}\nabla\Phi + \sqrt{2A}\mathbb{M}\boldsymbol{\zeta}(t) \tag{11}$$

Now make a change of variables (with unit Jacobian):

$$w_x = v_x + \frac{qB}{D}\partial_y\Phi \tag{12}$$

$$w_y = v_y - \frac{qB}{D}\partial_x\Phi \tag{13}$$

where D is determinant of $\mathbb{M}^{-1}$, i.e. $D = \Gamma^2 + q^2B^2$.

Then equation (61) becomes

$$\tau\dot{\boldsymbol{w}} = -\boldsymbol{w} - \frac{\tau\Gamma}{D}(v_k\partial_k)\nabla\Phi - \frac{\Gamma}{D}\nabla\Phi + \sqrt{2A}\mathbb{M}\boldsymbol{\zeta}(t) \tag{14}$$

Absorbing $\frac{\Gamma}{D}$ in potential Φ gives

$$\tau\dot{\boldsymbol{w}} = -\boldsymbol{w} - \tau(v_k\partial_k)\nabla\Phi - \nabla\Phi + \sqrt{2A}\mathbb{M}\boldsymbol{\zeta}(t) \tag{15}$$

Now rescaling the variable as in [4] gives:

$$\dot{\boldsymbol{w}} = -\frac{\boldsymbol{w}}{\sqrt{\tau}} - \sqrt{\tau}(v_k\partial_k)\nabla\Phi - \nabla\Phi + \sqrt{\frac{2A}{\sqrt{\tau}}}\mathbb{M}\boldsymbol{\zeta}(t) \tag{16}$$

where we have also done the re-scaling $\boldsymbol{B} \to \frac{\boldsymbol{B}}{\sqrt{\tau}}$.

So equation (60) in new variables and rescaled unit becomes

$$v_x = w_x - \frac{qB}{D}\partial_y\Phi \tag{17}$$

and

$$v_y = w_y + \frac{qB}{D}\partial_x\Phi \tag{18}$$

Now let's forget second term on RHS of equation (16) for a moment and show that closed form solution of fokker-plank equation corresponding to equations (16),(17) and (18) exists. After that, we will write a perturbative solution to account for that term. The Fokker-Planck equation is:

$$\partial_t P = -\nabla.\left(\boldsymbol{v}P\right) + \nabla_w.\left(\frac{\boldsymbol{w}}{\sqrt{\tau}}P\right) + \nabla_w.(\nabla\Phi P) + \frac{A}{\sqrt{\tau}D}\nabla_w^2 P \tag{19}$$

It's easy to check that

$$P = \exp\left(-\frac{\Phi}{T} - \frac{w^2}{2T}\right) \tag{20}$$

where

$$T = \frac{A}{D} \tag{21}$$

solves the above equation.

Let's define $Q = \frac{qB}{D}$ and $\mathcal{A}_i = \varepsilon_{ij}\partial_j\Phi$. Note that, by construction, divergence of $\boldsymbol{\mathcal{A}}$ is zero.

Equation (17) and (18) in these symbols are

$$v_x = w_x - Q\mathcal{A}_x \tag{22}$$

and

$$v_y = w_y - Q\mathcal{A}_y \tag{23}$$

Time derivative of canonical momentum($\boldsymbol{w}$) of a charged(Q) particle of mass m in magnetic field is given by

$$m\dot{w}_i = (w_j - Q\mathcal{A}_j)Q\partial_i\mathcal{A}_j \tag{24}$$

where $\boldsymbol{\mathcal{A}}$ is vector potential.

We can write second term on the RHS of equation (16) with the help of equations (22) and (23) as

$$\sqrt{\tau} v_k \partial_i (\partial_k \Phi) = -\sqrt{\tau}(w_j - Q\mathcal{A}_j)\partial_i(\varepsilon_{jl}\mathcal{A}_l) \tag{25}$$

We note here that (25) is almost like a Lorentz force contribution, in terms of the canonical momenta, with the difference arising out of the factor of ε_{jl} on the RHS.

Note The following section, till 5.3, is review material and not my original work. Part of the summer project was to learn this material as the basis for further research. In this part, I am basically summarising whatever I have learnt, on linear response theory, fluctuation-dissipation theorems and stochastic thermodynamics.

5 How far from equilibrium?

In order to understand the general problem of the previous section better, we would like to understand exactly to what extent our system is away from equilibrium. One tool that is commonly used to judge whether a system is away from equilibrium is to check if "detailed balance" holds. It is known from statistical physics that the probability of a process $A \to B$ happening is equal to that of $B \to A$ happening, when the system is in thermal equilibrium at a finite temperature. Mathematically,

$$P(A)W(A \to B) = P(B)W(B \to A)$$

with $P(i)$ representing the probability of the system being in state i, and $W(i \to j)$ being the *transition probability* of the system going from state i to state j. This is what is meant by detailed balance being satisfied.

In non-equilibrium systems, interestingly, this balance of probabilities is broken, by some kind of driving mechanism present in the system, for example molecular motors present inside cells, or, as a simplified model, noisy forces which have a finite memory. This shows up as probability currents as well as currents in configuration space, even in *steady state*.

Besides this somewhat qualitative idea of the breaking of a balance, we are also interested in quantifying the non-equilibrium-*ness* of our system. In this regard, one plausible approach to take is to see if some kind of generalised version of what are called "fluctuation-dissipation" theorems can be formulated for our problem. This idea has been explored in a recent paper by Fodor et al[4]. In order to explore this idea, it is important to make a digression to linear response theory.

5.1 Linear Response

Linear response theory is an extremely general framework, which predicts the behaviour of a system under the influence of small, time-dependent perturbations, upto first order in the perturbation (hence "linear"). For example, if we create a temperature gradient in a metal bar, heat begins to flow, the rate at which it flows being determined by its material properties. In general, the way a system responds to an external perturbing "force"(I have put it in quotes because this perturbation need not dimensionally be an actual force) is determined by something called a "response function". It turns out that this response function is determined solely by certain correlations, evaluated at *thermal equilibrium*, although due to the external perturbation, the system is not at equilibrium at all times.

We can attempt to understand this phenomenon intuitively as follows. At thermal equilibrium, there are random fluctuations constantly occurring throughout the system, which get damped down by the relevant damping mechanism, e.g. frictional damping, thereby dissipating energy. Thus, at equilibrium, there is some kind of a balance being constantly maintained by the fluctuations and dissipation. When an external perturbation is applied, the first order effects of it are somewhat like thermal fluctuations, and hence the way the system responds to such perturbations must be like the way it dissipates energy from those fluctuations. The latter effect, however, is completely determined by the equilibrium probability distribution of the system. Motivated by this, we can get an intuitive feel for why the response of the system to small perturbations could be determined by correlation functions, which essentially quantify thermal fluctuations.

Let us now look at how exactly external forces can be incorporated into a given system. In classical mechanics we have Newton's law, which gives rise to equations of the form

$$m\ddot{x}_i = f_i(\boldsymbol{x}, \dot{\boldsymbol{x}}) \tag{26}$$

$\boldsymbol{f}$ being the net force on the particle.

Now, if we want to incorporate some time dependent external force $\boldsymbol{F}(t)$, we can re-write (26) as

$$m\ddot{x}_i = f_i(\boldsymbol{x}, \dot{\boldsymbol{x}}) + F_i(t) \tag{27}$$

It is to be noted that the external force doesn't have any space dependence, and its time-dependence is something that we get to decide.

In quantum mechanics, we incorporate the perturbations in the Hamiltonian of the system. Let H_0 be the equilibrium Hamiltonian of the system, the total Hamiltonian H_{tot} after introduction of the perturbation can be written as

$$\hat{H}_{tot}(t) = \hat{H}_0 - F(t)\hat{A} \tag{28}$$

where $\hat{A}$ is the observable conjugate (w.r.t. energy) to the perturbation F(t). For example, for a spatially constant perturbing force F, the observable would be position, i.e. $\hat{A} \equiv \hat{x}$.

A similar formulation is also possible in classical mechanics, if we assume that the system is a Hamiltonian one. We would need to replace the operator $\hat{A}$ in (28) with the function $A(q, p)$, a function of the canonical positions and momenta $\{q, p\}$. We write down the time dependent Hamiltonian of the system as

$$H_{tot}(t) = H_0 - A(q_0, p_0)F(t) \tag{29}$$

with $\{q_0, p_0\}$ representing the initial position and momenta, respectively. For the rest of this section, we stick to the quantum mechanical notation.

Let us make a short digression to describe what we mean by expectation value, in both the quantum mechanical and the classical version of the system. A quantum system in a certain state ket $|\psi\rangle$ has a density matrix defined on it as $\rho = |\psi\rangle\langle\psi|$. For a classical system, the phase space density ρ would be something like a delta function at its time-dependent location at each point of time, since for a given initial condition, the classical trajectory is fully determined. Then, the expectation value of any observable $\hat{A}$, in quantum as well as classical cases, we would have $\langle A\rangle = Tr(\rho A)$, where $Tr(..)$ represents a sum over all states of the ensemble. It is a simple task to check that this definition of expectation value matches the more commonly known quantum mechanical expression $\langle\psi|A|\psi\rangle$. Note here that we can either interpret the density matrix as evolving with time, keeping the operator constant (Schrodinger picture), or the other way around (Heisenberg picture), the expectation value is unaffected.

Under the influence of the perturbing force, let the expectation value of an observable B go from $\langle B\rangle_{eq}$ to $\langle B(t)\rangle$, so that the difference is $\langle \delta B(t)\rangle$. By definition,

$$\begin{aligned}
\langle B\rangle_{\rho(t)} &= Tr(\rho(t)B) \\
&= Tr((\rho_{eq} + \delta\rho(t))B) \\
&= \langle B\rangle_{eq} + \langle B\rangle_{\delta\rho(t)}
\end{aligned} \tag{30}$$

Therefore, we conclude that $\langle \delta B(t)\rangle = \langle B\rangle_{\delta\rho(t)}$ In quantum mechanics, as well as classical mechanics, there are time evolution equations for ρ which can be written together as

$$\partial_t \rho = -i\mathcal{L}\rho \tag{31}$$

where the Liouville operator $\mathcal{L}$ is defined as

$$\mathcal{L}\rho = -i(\rho, H) \tag{32}$$

with the notation (A, B) representing the Poisson bracket $\{A, B\}$ in classical mechanics, and the commutator $\dfrac{[A, B]}{i\hbar}$ in quantum mechanics. In presence of the perturbation we can split the Liouville operator as $\mathcal{L} = \mathcal{L}_{eq} + \mathcal{L}_{pert}$, and hence we can use (31) as

$$\begin{aligned}
\frac{\partial\rho}{\partial t} &= -i(\mathcal{L}_{eq} + \mathcal{L}_{pert})(\rho_{eq} + \delta\rho(t)) \\
&\approx -i(\mathcal{L}_{eq}\delta\rho(t) + \mathcal{L}_{pert}\rho_{eq})
\end{aligned} \tag{33}$$

where we have used the fact that, by definition of equilibrium, $\dfrac{\partial\rho_{eq}}{\partial t} = -i\mathcal{L}_{eq}\rho_{eq} = 0$, and that for small perturbations, the term $\mathcal{L}_{pert}\delta\rho(t)$ is negligible since it is second order in the perturbation. The solution to equation (33) is

$$\delta\rho(t) = -\int_{-\infty}^{t} dt' F(t')(A(t'-t), \rho_{eq}) \tag{34}$$

where $A(t'-t)$ represents evolution via the equilibrium Hamiltonian H_0. Using (34) and the definition of $\langle \delta B(t) \rangle$, we derive

$$\langle \delta B(t) \rangle = -\int_{-\infty}^{t} dt' F(t') \phi_{AB}(t-t') \tag{35}$$

where we have introduced the notation

$$\phi_{AB}(\tau) = Tr[(A(0), B(\tau))\rho_{eq}] = \langle (A(0), B(\tau)) \rangle_{eq} \tag{36}$$

ϕ_{AB} is known as the *response function.* Note that $\rho_{eq} = e^{-\beta H_0}$, where $\beta = \frac{1}{k_B T}$, according to canonical distribution at temperature T.

This expression for the response function can be further simplified into a simple canonical correlation function, due to Kubo, as

$$\phi_{AB}(\tau) = \beta \langle \dot{A}(0); B(\tau) \rangle_{eq} \tag{37}$$

where the notation $\langle X; Y \rangle_{eq}$ denotes

$$\langle \dot{X}(0) Y(\tau) \rangle_{eq}$$

for classical mechanics, and

$$\frac{1}{\beta} \int_0^{\beta} d\lambda \langle e^{\lambda H_0} X e^{-\lambda H_0} Y \rangle$$

for quantum mechanics.

We are now in a position to appreciate the statement of the fluctuation-dissipation theorem. In the next section, we look at one of the first statements of this theorem, due to Kubo [8].

5.2 Fluctuation-dissipation theorem

A quantity related to the response function is the generalised susceptibility χ_{AB}, which is nothing but the one-sided Fourier transform (a.k.a. Fourier-Laplace transform) of ϕ_{AB} defined as

$$\chi_{AB}(\omega) = \int_0^{\infty} d\tau e^{-i\omega\tau} \phi_{AB}(\tau) \tag{38}$$

It can be shown that the power dissipated by the system is proportional to the imaginary part of χ. This fact has been derived in reference [9]. In his famous paper on fluctuation dissipation theorem [8], Kubo states the following form of equation (38) as one of the "exact expressions of the fluctuation-dissipation theorem".

$$\chi_{AB}(\omega) = \beta \int_0^\infty d\tau e^{-i\omega\tau} \langle \dot{A}(0); B(\tau) \rangle_{eq} \tag{39}$$

Taking the imaginary part of $\chi_A B$ would give a quantitative measure of the dissipation of the system, which is related to the fluctuations of the system at equilibrium via (39). This is the basic idea behind the so-called fluctuation dissipation, or fluctuation response theorem.

Let us now derive a fluctuation-dissipation theorem for a system described by a generalised Langevin equation

$$m\frac{d}{dt}u(t) = -\int_{-\infty}^{t} dt'\gamma(t-t')u(t') + K(t) + R(t) \tag{40}$$

where K is a time-dependent external force (assumed to be a "small" perturbation), and R(t) is a random force, with the statistical properties

$$\langle R(t) \rangle = 0 \tag{41}$$

and for $t_1 < t_2$ (by causality)

$$\langle u(t_1)R(t_2) \rangle = 0 \tag{42}$$

We can define a susceptibility $\mu(\omega)$ for this system by applying linear response theory, with $A = x$ and $B = u \equiv \dot{x}$. Then we have, using (39)

$$\mu(\omega) = \beta \int_0^\infty d\tau e^{-i\omega\tau} \langle u(0); u(\tau) \rangle_{eq} \tag{43}$$

Now, we use equation (40) to write an expression for $\mu(\omega)$ in terms of γ. Taking ensemble average of (40),

$$m\frac{d}{dt}\langle u(t) \rangle = -\int_{-\infty}^{t} dt'\gamma(t-t')\langle u(t') \rangle + K(t) \tag{44}$$

Now, assuming a periodic force for $K(t)$ (angular frequency ω), we get

$$\langle u(t) \rangle = Re\left(\mu(\omega)K_0 e^{-i\omega t}\right) \tag{45}$$

where $\mu(\omega) = \dfrac{1/m}{i\omega + \gamma[\omega]}$, $\gamma[\omega]$ being the Fourier-Laplace transform of $\gamma(t)$. This definition of the mobility $\mu(\omega)$ must agree with that defined by (43). We can intuitively see that this will lead us to the fluctuation-dissipation theorem for this system. Indeed, as shown by Kubo[8], we have

$$m\gamma[\omega] = \beta \int_0^\infty \langle R(t_0); R(t_0+t) \rangle e^{-i\omega t} dt \tag{46}$$

where the r.h.s. is a measure of the equilibrium fluctuations of the random force, and the l.h.s. is the Fourier-Laplace transform of the friction memory kernel $\gamma(t)$, which is nothing but a quantitative measure of dissipation.

Thus, equation (46) is a fluctuation-dissipation theorem for our example system, described by the generalised Langevin equation (40).

In order to talk about generalised fluctuation-dissipation theorems in non-equilibrium systems, we need the language of stochastic thermodynamics. In the following section, we look at some of the basic ideas of the same.

5.3 Stochastic thermodynamics

One of the earliest papers on this topic was by Sekimoto[10], where the author formulates an equivalent of the thermodynamic first law for individual trajectories, defining the quantities work and heat appropriately. The system is assumed to be in contact with a heat bath, having a fixed finite temperature, and hence described by a Langevin equation as follows.

$$\gamma \frac{dx}{dt} = -\frac{dU}{dx} + \xi(t) \tag{47}$$

with ξ satisfying the statistical properties

$$\langle \xi(t) \rangle = 0, \quad \langle \xi(t)\xi(t') \rangle = 2\gamma T \delta(t-t'). \tag{48}$$

Now we re-write equation (47), after multiplying throughout by dx, to get

$$0 = -\left(-\gamma \frac{dx}{dt} + \xi(t)\right) dx + \frac{dU}{dx} dx \tag{49}$$

We identify the first term on the r.h.s. of (49) as the heat dQ discarded by the system to the heat bath, since $-\left(-\gamma \frac{dx}{dt} + \xi(t)\right)$ is the reaction force of the system on the heat bath. Note that dQ is a stochastic quantity which is determined by the *particular realisation* of noise that we observe.

We generalise this a bit to incorporate the dependence of the potential of the system U on an external parameter a as well. Under such a circumstance, we can write the following

$$\frac{\partial U}{\partial a} da = dQ + dU \tag{50}$$

We can identify the l.h.s. of (50) as the work dW done *by the external system*, by conservation of energy in mechanics. Thus,

$$dW = dQ + dU \tag{51}$$

which at this point is just a relation between stochastic quantities. When we take ensemble averages on both sides of (51), we derive

$$\left\langle \frac{\partial U}{\partial a} \right\rangle da = \langle dQ \rangle + \langle dU \rangle \tag{52}$$

Now that we have framed an analogue of the first law in the stochastic framework, we would like to define *entropy* for a stochastic trajectory.

Entropy, in equilibrium thermodynamics, is an ensemble property. It is defined as

$$S = k_B T \ln \Omega \tag{53}$$

where Ω is the multiplicity of the ensemble, i.e. the number of accessible microstates for the system in a particular ensemble. Thus, it is intuitively slightly hard to visualise what entropy for a trajectory would mean. At the same time, however, the fluctuation theorem[11] quite generally talks about the probabilities of entropy-producing and entropy-destroying trajectories. This discussion is carried out in reference [12]. Here we are reproducing part of that.

A common definition of non-equilibrium Gibbs entropy (defined for an ensemble) is

$$S(\tau) \equiv -\int p(x,\tau) \ln p(x,\tau) \tag{54}$$

This motivates a definition for entropy defined for a particular stochastic trajectory $x(\tau)$ as

$$s(\tau) = -\ln p(x(\tau),\tau) \tag{55}$$

where the probability distribution $p(x,\tau)$ is determined by the Fokker-Planck equation describing the system. It is clear that $S(\tau) = \langle s(\tau) \rangle$ where the ensemble average has been taken over different realisations of the stochastic trajectory.

Let us consider the following Langevin equation, which is a slight modification of equation (47)

$$\dot{x} = \mu F(x,\lambda) + \xi(t) \tag{56}$$

where λ is an external parameter, and $xi(t)$ is a Gaussian white noise term. The rate of change of the entropy of the system, as defined in (55) is

$$\begin{aligned} \dot{s}(\tau) &= -\frac{\partial_\tau p(x,\tau)}{p(x,\tau)}\Big|_{x(\tau)} - \frac{\partial_x p(x,\tau)}{p(x,\tau)}\Big|_{x(\tau)} \dot{x} \\ &= -\frac{\partial_\tau p(x,\tau)}{p(x,\tau)}\Big|_{x(\tau)} + \frac{j(x,\tau)}{Dp(x,\tau)}\Big|_{x(\tau)} \dot{x} - \frac{\mu F(x,\lambda)}{D}\Big|_{x(\tau)} \dot{x} \end{aligned} \tag{57}$$

Now we are going to define a rate of change of the entropy of the medium $s_m(\tau)$ as follows

$$\dot{q}(\tau) = F(x,\lambda)\dot{x}, \quad \dot{s}_m(\tau) \equiv \frac{\dot{q}(\tau)}{T} \tag{58}$$

The first of the above two equalities is due to Sekimoto's definition of the stochastic heat[10].

Thus, we can now define a total entropy which evolves as follows

$$\begin{aligned} \dot{s}_{tot}(\tau) &= \dot{s}_m(\tau) + \dot{s}(\tau) \\ &= -\frac{\partial_\tau p(x,\tau)}{p(x,\tau)}\Big|_{x(\tau)} + \frac{j(x,\tau)}{Dp(x,\tau)}\Big|_{x(\tau)} \dot{x} \end{aligned} \tag{59}$$

This is one of the main results of the paper [12] by Seifert. Such a definition of entropy for the particle/system defined by the Langevin equation (56) has been shown to give correct balance equation for the ensemble Gibbs entropy. A fluctuation theorem can be devised using this definition of stochastic entropy, which we are not going to go into in this report.

5.4 Back to the problem

Now that we have sufficient machinery, we can talk about entropy production rates, and generalised fluctuation dissipation theorems for the problem at hand.

Let us remind ourselves of the equations of motion of our system.

$$\dot{\boldsymbol{x}} = \boldsymbol{v} \tag{60}$$

$$\tau\dot{\boldsymbol{v}} = -\tau(v_k\partial_k)\mathbb{M}\nabla\Phi - \boldsymbol{v} - \mathbb{M}\nabla\Phi + \sqrt{2A}\mathbb{M}\boldsymbol{\zeta}(t) \tag{61}$$

In reference [4], a calculation for the entropy production rate σ is carried out, which is being partly reproduced here. In terms of probabilities of trajectories,

$$\sigma = \lim_{t\to\infty} \frac{1}{t} \ln \frac{P}{P_R} \tag{62}$$

where P and P_R denote the probability weights associated with a given trajectory and its time-reversed counterpart, respectively. We use the path integral approach to Langevin dynamics in order to interpret these probabilities.

The probability of a trajectory $\{\boldsymbol{x}(t)\}$

$$P(\{\boldsymbol{x}(t)\}) \propto \exp(-S[\{\boldsymbol{x}(t)\}]) \tag{63}$$

where the action S is given by

$$S = \frac{1}{4\tilde{A}} \int_0^t dt' \left(\tau\dot{\boldsymbol{v}} + \tau(v_k\partial_k)\mathbb{M}\nabla\Phi + \boldsymbol{v} + \mathbb{M}\nabla\Phi\right)^2 \tag{64}$$

where $\tilde{A} = \frac{A}{D}$, with D defined as in section 4. Under time-reversal, we perform the transformations

$$\begin{aligned} t_R &= -t \\ \boldsymbol{x}_R(t) &= \boldsymbol{x}(-t) \\ \boldsymbol{v}_R(t) &= -\boldsymbol{v}(-t) \end{aligned}$$

With this idea, we can calculate the rate of production of entropy as $\sigma \sim \frac{\delta S}{t}$, where δS is the difference between the entropy of the forward and backward actions.

The road ahead

In the course of this project, we have seen how an over-damped active Brownian particle in a magnetic field is a non-equilibrium system which can be studied from various perspectives, revealing certain interesting characteristics. There is more that can be done with this problem, in particular regarding the quantification of the question “how far from equilibrium”, which was discussed in the previous section of this report. This can be carried forward from the calculation of the rate of entropy production.

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